

Phetolo Tshukudu

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PROFESSIONAL SUMMARY

Final-year BSc Mathematical Sciences student at the University of Johannesburg, majoring in Computer Science and Mathematical Statistics, with a strong foundation in software engineering, backend development and data analysis. Hands-on experience building Java/Spring Boot REST APIs and Python data-analysis applications, with practical experience in relational databases, object-oriented programming, automated testing and statistical analysis. Interested in graduate and junior software engineering, backend development and data engineering opportunities where I can apply strong programming, analytical and problem-solving skills to build reliable, data-driven systems.

EDUCATION

University of Johannesburg

2023 - Present

BSc. Mathematical Sciences in Computer Science and Mathematical Statistics

Relevant coursework

Object-Oriented Programming • Data Structures & Algorithms • Database Systems • Software Engineering • Computer Networks • Operating Systems • Mathematical Statistics • Statistical Modelling • Regression Analysis

Schooling /Career Highlights

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- High school learner council
 - Academic award in Mathematics
 - UJ UJenius Certificate

SKILLS

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- **Languages:** Java, Python ,SQL, R
 - **Backend:** Spring Boot, Spring Security, REST APIs, JWT, JPA/Hibernate
 - **Python & Data:** Pandas, Data Cleaning, Data Analysis, Statistical Analysis, Dataclasses
 - **Databases:** MySQL, SQLite, Relational Database Design
 - **Testing:** Pytest, JUnit 5, Mockito
 - **Development Tools:** Git, GitHub, Maven, Eclipse IDE, Postman, MySQL Workbench
 - **Core Computer Science:** Data Structures & Algorithms, Graph Algorithms, Database Systems, Software Engineering
 - **Statistics & Data:** Statistical Modelling, Regression Analysis, Data Analysis, Data Interpretation & Visualisation, Model Evaluation & Validation

WORK EXPERIENCE (PART TIME WORK)

MATHEMATICS & PHYSICAL SCIENCES TUTOR Ghetto Hero Saturday School | Jun 2023 – Jul 2023

- Tutored Grade 10–12 learners in Mathematics and Physical Sciences.

- Explained complex concepts using structured, accessible problem-solving approaches.
- Supported learners in developing mathematical reasoning and problem-solving skills.

LEADERSHIP EXPERIENCE

TEAM LEADER — FREE SANITARY PAD & TOILETRY PROJECT Bridge to Employment Programme, NICDAM

- Led a team in organising a community project focused on providing sanitary products and toiletries.
- Coordinated team responsibilities and project activities to support successful project delivery.
- Engaged potential sponsors to seek donations and support for the initiative.
- Developed practical experience in leadership, teamwork, communication and project coordination.

FINANCIAL DATA ANALYZER

Python | Pandas | Pytest | Dataclasses | Statistics | Git/GitHub

- Built a Python-based financial analysis system for processing, cleaning and analysing transaction data.
- Implemented object-oriented domain models using Python classes and @dataclass, with separate services for transactions, budgeting and statistical analysis.
- Added automated tests using pytest, with 16 passing tests covering core business logic and edge cases.
- Used Pandas for CSV data processing and implemented spending analysis, financial statistics and budget tracking.
- Structured the application into models, services, data processing and tests to improve maintainability and separation of responsibilities.

PERSONALFINANCE API — Secure Financial Management Backend Java | Spring Boot | Spring Security | JWT | MySQL | JPA/Hibernate | JUnit 5 | Mockito | Maven

- Developed a RESTful backend for managing users, accounts, transactions and personal budgets using Java and Spring Boot.
- Implemented Spring Security and JWT-based authentication/authorisation to secure protected API endpoints.
- Designed a relational persistence layer using JPA/Hibernate and MySQL, including entity relationships and repository-based data access.
- Applied layered architecture using controllers, services, repositories and DTOs to separate API, business and persistence responsibilities.
- Implemented validation and exception handling to improve API reliability and provide consistent error responses.
- Developed unit tests using JUnit 5 and Mockito to verify service-layer business logic and repository interactions.

WILDFIRE SIMULATOR Java | JavaFX | Data Structures & Algorithms | Graph Algorithms | Dijkstra's Algorithm | Git

- Collaborated on a Java application that simulates wildfire propagation across a terrain and assists users in identifying safer escape routes.
- Implemented graph-based pathfinding using Dijkstra's algorithm to determine routes based on terrain/fire conditions.
- Applied priority queues and custom data structures to model and process wildfire propagation.
- Contributed to the JavaFX interface and integrated algorithmic components into the application.
- Used Git-based collaboration and feature branches during team development.